P-192 batched three-coordinate labelled transition

Translate three coordinates with tracked labels

One batch inversion for 3B additions

Lockstep tables and signed-digit ladder

102 further inversion layers; exceptions recorded

Canonicalize and update labels

32 checked transitions; eight independently recounted cost vectors

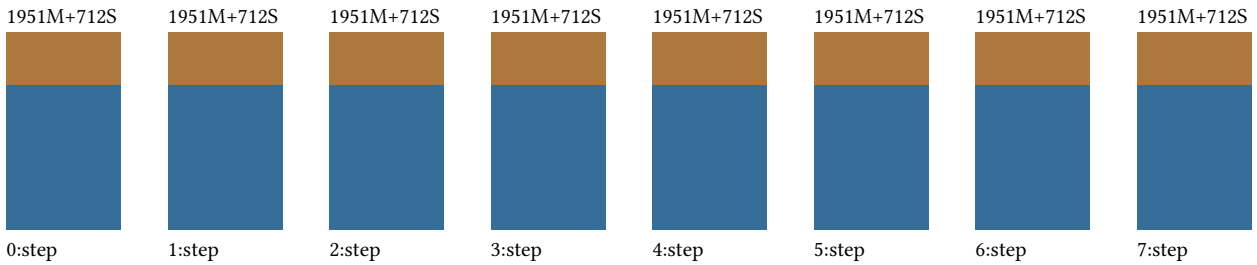
Derived construction. Use the power-two fifth root from round46 and store R, lambda R, lambda^2 R with scalar labels (a,b), where R=aG+bQ. Partition the canonical first point into 16 deterministic table entries. Translate the three stored coordinates, evaluate the selected triple signed-digit relation in lockstep across walkers, reconstruct the fifth point by the sum-zero identity, and canonicalize by minimum x and sign. Rotation j and sign s multiply both updated scalar labels by s*lambda^j modulo r.

Derived generic field cost. For B walkers with no identity, cancellation or accidental doubling in an addition layer, the complete transition costs (565B-309)M +178B S +103I +(633B+3s)A +130B K, where s counts negative canonical signs. M, S, I and A are field multiplication, squaring, inversion and addition/subtraction; K counts multiplication by a small constant. Scalar-label arithmetic and hashes are separate. Input states and the 16-entry translation table are initialized outside this transition cost.

Derivation. Translation uses one batch of 3B additions. Three input-table doublings share one layer of 3B points; nine table additions share three such layers. The digit ladder has 62 doubling and 33 addition layers of B points, followed by three sum layers. Thus there are 103 inversion layers, 65B doublings and 48B ordinary additions. A batch of n point operations costs (5n-3) multiplications. Doublings contribute two squarings, two small-constant products and five field additions; ordinary additions contribute one squaring and six field additions. Eighteen negative lookups, the fifth-point negation, and canonical sign selection add 20B field subtractions, with three more for each negative output sign.

Exact fixture checks. Four walkers take eight complete steps using the frozen public secret 20261107, deterministic initial labels and translation table. Every step records 1951M, 712S, 103I and 520K; additions range from 2532 to 2544 according to sign choices. The separate checker reconstructs all eight cost vectors from operation branches and checks all 32 scalar-label transitions and direct three-point encodings. All checks pass locally. There are zero repeat records in these 32 transitions; this small deterministic fixture gives no mixing or cycle-rate estimate. [pcc:unverified]

Round47: field counts per four-walker transition



Blue: field multiplications. Brown: field squarings.

Each step also uses 103 inversions; setup and scalar labels separate.

Scope. The arithmetic code handles exceptional inputs, but this round validates the generic layer formula and the listed nonexceptional walk fixtures. Identity, cancellation, planted recovery and equivalent encodings have controls in rounds44 and46; a unified adversarial replay of the new lockstep transition remains open. The recorded group schedule has become an executable labelled walk component, with setup exclusions explicit.

Next construction. Investigate a transition invariant or update rule that avoids reconstructing the missing orbit coordinates on every step. Simply delaying canonicalization does not supply an orbit walk: the partition and transition must commute with multiplication by lambda. A candidate must exhibit that equivariance and account for its invariant evaluation, rather than assume the earlier canonicalization cost disappears.